Problem Statement 1

Create the following four tables with appropriate Primary Keys and Foreign Keys, and insert 10 records into each table.

Customer (cust_id, cust_name, city, phone)
Branch (branch_id, branch_name, branch_city)
Account (acc_no, cust_id, branch_id, balance)
Loan (loan_no, cust_id, branch_id, amount)

1. Display all customers in alphabetical order of their names.
2. Display the names and cities of customers whose names start with the letter 'P'.
3. Display customers who live in Pune or Mumbai and whose phone number is not NULL.
4. Display all customers whose phone number is NULL.
5. Display the average account balance for each branch.
6. Display the branch-wise average account balance and show only those branches where the average balance is greater than ₹40,000.
7. Display each account number along with its balance after adding 10% interest. Do not modify the actual balance.
8. Update the phone number of a specific customer using the customer's name.
9. Delete all customer records for customers who live in Pune.
10. Display customers who either have an account or have a loan using UNION.

Problem Statement 2

Create the following tables with appropriate Primary Keys and Foreign Keys, and insert 10 records into each table.

Students (student_id, name, city)
Enrolments (enroll_id, student_id, course, marks)

1. Display student names along with their course names and marks.
2. Display all students along with their course names and marks, including students who are not enrolled in any course.
3. Display all enrollments along with the corresponding student names.
4. Display all possible combinations of students and courses.
5. Display students whose marks are greater than the average marks of all enrollments .
6. Display students who have obtained marks greater than 80
7. Display students who are enrolled in the DBMS course
8. Display the student ID(s) who have obtained the highest marks
9. Create a view named student_course_view displaying student names, course names, and marks. Display all records from the view and update the marks of a student through the view.
10. Create an index on the name column of the Students table and a unique index on the enroll_id column of the Enrollments table

Problem Statement 3

Create the following four tables with appropriate Primary Keys and Foreign Keys, and insert 10 records into each table.

Students(student_id, name, city)
Enrollments(enroll_id, student_id, course, marks)

1. Display student names and the courses in which they are enrolled.
2. Display all students and their course details, including students who are not enrolled in any course.
3. Display the names of students enrolled in DBMS along with their marks.
4. Display enrollments that do not have a matching student record.
5. Display students who have obtained marks less than the average marks of all enrollments.
6. Display students who have obtained marks equal to the highest marks.
7. Display students who belong to Pune.
8. Display students enrolled in Python.
9. Create a view named course_average_view displaying course names and their average marks. Display all records from the view.
10. Create an index on Students.city and a unique index on Students.student_id.

Problem Statement 4

Create the following tables and insert 10 records.

Stud_Marks (rollno,name, total_marks) Result (Roll, Name, Class)

Write a Stored Procedure namely proc_Grade for the categorization of student. If marks scored by students in examination is <=1500 and marks>=990 then student will be placed in distinction category if marks scored are between 989 and 900 category is first class, if marks 899 and 825 category is Higher Second Class.

Problem Statement 5

Write a PL/SQL function code block to calculate the area of a circle for a value of radius varying from 5 to 9. Store the radius and the corresponding values of calculated area in an empty table named areas, consisting of two columns, radius and area.

Problem Statement 6

Create an Employee table with attributes Emp_ID, Emp_Name, Department, and Salary. Write a PL/SQL program using a Cursor to fetch and display the details of all employees whose salary is greater than ₹30,000. Use the cursor to retrieve records one by one and display the employee information.

Problem Statement 7

Create following tables and inert 10 records

Library (book_id, book_name, author, category, date, operation)
Library_Audit (book_id, book_name, author, category, date, operation)

Write a database trigger on Library table. The System should keep track of the records that are being updated or deleted. The old value of updated or deleted records should be added in Library_Audit table

Problem Statement 8

Write a program to implement MySQL/Oracle database connectivity with any front end language to implement Database navigation operations (add, delete, edit etc.)

Problem Statement 9

Write a program to implement MongoDB database connectivity with any front end language to implement Database navigation operations (add, delete, edit etc.)

Problem Statement 10

Consider following Bank database schema and solve given queries:

Account(Acc_no, branch_name,balance) branch(branch_name,branch_city, assets)

customer(cust_name,cust_street,cust_city) Depositor(cust_name,acc_no)

Loan(loan_no,branch_name,amount) Borrower(cust_name,loan_no)

1. Create above tables with appropriate constraints like primary key, foreign key, not null etc. with suitable data

2. Add customer phone number in Customer table.
3. Delete phone number attribute from Customer table.
4. Find the names of all branches in loan relation.
5. Find all customers who have a loan from bank. Find their names,loan_no and loan amount
6. List all customers in alphabetical order who have loan from Akurdi branch.
7. Find average account balance at Chinchwad branch.
8. Find no. of depositors at each branch
9. Find the total loan amount for each branch.
10. Find the branches where the average loan amount is greater than 5,00,000.

Problem Statement 11

Create the above tables with appropriate constraints such as primary key, foreign key, NOT NULL, etc., and insert suitable data.

Account(Acc_no, branch_name, balance)

Branch(branch_name, branch_city, assets)

Customer(cust_name, cust_street, cust_city)

Depositor(cust_name, acc_no)

Loan(loan_no, branch_name, amount)

Borrower(cust_name, loan_no)

1. Modify the assets attribute of the Branch table to Property.
2. Find all loan numbers for loans made at Akurdi branch with a loan amount greater than 12,000.
3. Find the average account balance at each branch.
4. Find the branches where the average account balance is greater than 12,000.
5. Find the number of tuples in the Customer relation.
6. Calculate the total loan amount given by the bank.
7. Delete all loans with a loan amount between 1,300 and 1,500.
8. Find the customers whose street address is not available.
9. Find the branches where the branch city is available.

10. Find the names of customers whose names start with the letter 'A'.

Problem Statement 12

Consider the employee collection with the following attributes:

employee(empid, ename, dept, salary)

Create a database named ICEM and an employee collection in **MongoDB.** Insert suitable data and perform the following queries.

1. Display the information of all employees.

2. Display the information of employees working in the Computer department.

3. Display the information of employees working in the Computer or IT department using a logical operator.

4. Display the information of employees whose salary is greater than ₹30,000.

5. Display the information of employees whose salary is greater than or equal to ₹40,000 and who work in the Computer department.

6. Create an index on the ename field and drop the index created on the ename field in the employee collection..

7. Display the information of employees whose salary is between ₹20,000 and ₹40,000.

8. Display the information of employees whose department is not IT.

9. Display the total salary paid to employees in each department.

10. Display the average salary of employees in each department.

Problem Statement 13

Consider the student collection with the following attributes:

student(roll_no, sname, dept, marks)

Create a database named Student and a student collection in **MongoDB.** Insert suitable data and perform the following queries.

1. Display the information of all students.

2. Display the information of students belonging to the Computer department.

3. Display the information of students belonging to the Computer or IT department using a logical operator.

4. Display the information of students whose marks are greater than 70.

5. Display the information of students whose marks are greater than or equal to 80 and who belong to the Computer department.

6. Display the information of students whose marks are less than 40 or who belong to the IT department.

7. Display the information of students whose marks are between 40 and 80.

8. Display the information of students whose department is not IT.

9. Display the total marks of students in each department.

10. Create an index on the sname field and drop the index created on the sname field in the student collection.

Problem Statement 14

Create the following four tables with appropriate Primary Keys and Foreign Keys, and insert 10 records into each table.

Customer (cust_id, cust_name, city, phone)
Branch (branch_id, branch_name, branch_city)
Account (acc_no, cust_id, branch_id, balance)
Loan (loan_no, cust_id, branch_id, amount)

1. Display all customers who live in Pune or Mumbai, arranged in alphabetical order of their names.
2. Display the names of customers whose names contain the letter 'a'.
3. Display all customers whose phone number is NOT NULL.
4. Display customers who live in Pune and whose names start with 'P', or customers who live in Mumbai.
5. Display the maximum loan amount and the average loan amount for each branch.
6. Display branches whose total loan amount is greater than ₹1,00,000.
7. Display each account number along with its balance after adding ₹500. Do not modify the actual balance.
8. Change the branch of account number 2005 to a different branch.
9. Delete all customers whose phone number is NULL.
10. Display:a) Customers who have both an account and a loan using INTERSECT.

    b) Customers who have an account but no loan using MINUS.